

Retail Sales BI Project – One Page Summary

Project Title: Retail Sales Performance Analysis

Dataset: Superstore Sales Dataset (2023) – includes Order Date, Customer, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit

Case Study Objective:

Analyze retail sales performance across categories and customers; identify high, medium, and low-profit orders; and perform customer segmentation to guide business decisions.

1. Power BI Dashboard

- **Data Preparation:** Cleaned duplicates, corrected data types, handled missing values, created Profit Margin and Month Number columns.
 - **KPIs:** Total Sales, Total Profit, Total Quantity, Profit Margin %, Total Discount.
 - **Visuals (6+):**
 - Sales by Category (Bar Chart)
 - Sales Trend Over Months (Line Chart)
 - Sales by Segment (Donut Chart)
 - Discount vs Profit (Scatter Chart – each dot represents one order, size by Sales)
 - Profit Contribution by Category (Donut Chart)
 - Total Profit VS Target (Gauge visual)
 - **Interactivity:** Slicers for Month, Category.
 - **Insights:** Peak sales appear in November–December, the East region is the most profitable, and higher discounts are clearly associated with lower or negative profit. Profit contribution analysis highlights which categories truly drive profitability, not just sales volume.
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2. Machine Learning Models (RapidMiner)

A. Clustering (K-Means)

- **Features:** Sales, Profit, Quantity, Discount
- **Clusters:** 3 groups – High, Medium, Low-value customers
- **Insights:**
 - Cluster 1: High-value → high sales & profit
 - Cluster 2: Medium-value → moderate sales & profit
 - Cluster 3: Low-value → low sales & profit

- **Business Application:** Targeted marketing, loyalty programs, and upselling strategies.

B. Regression (Linear Regression)

- **Target:** Profit
- **Features:** Sales, Quantity, Discount
- **Performance Metrics:** RMSE, MAE, R^2
- **Insights:** Sales and quantity positively affect profit; high discounts reduce profit. Useful for predicting profitability of new orders.

C. Classification (Decision Tree)

- **Target:** Profit_Class (High, Medium, Low) – created from Profit column
- **Features:** Sales, Quantity, Discount
- **Performance Metrics:** Accuracy, Precision, Recall, F1-score
- **Insights:** Helps identify high, medium, and low-profit orders; supports operational decisions and promotions.

Conclusion:

The Power BI dashboard provides clear, interactive insights into sales trends, profitable regions, and high-performing categories. The ML models (clustering, regression, classification) enable data-driven customer segmentation, profit prediction, and order classification, supporting strategic decisions for maximizing revenue and efficiency.